

Digital Logic Design ECE 3441 – Fall 2025

Lab 8:

Team:

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Design of a Traffic Controller

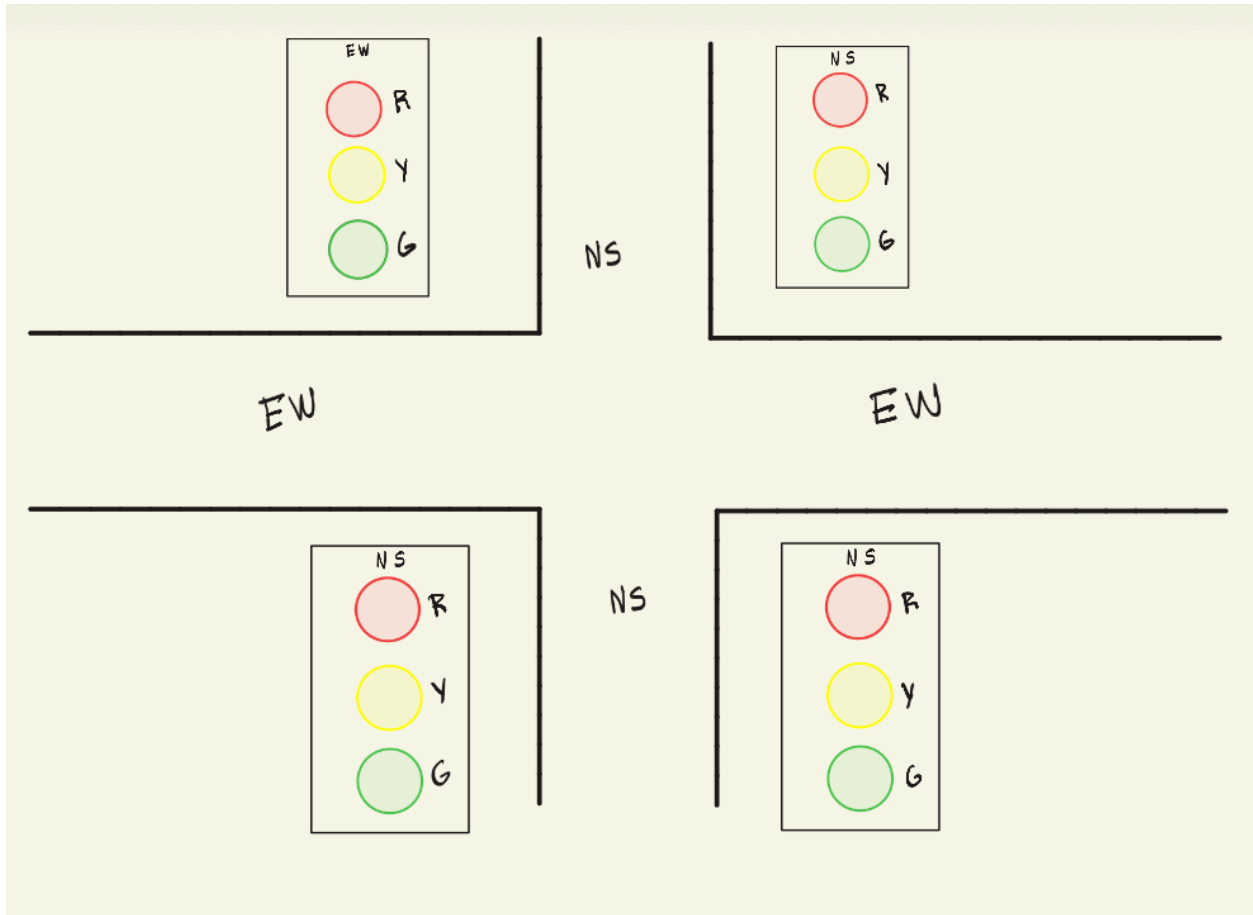
Overview

The purpose of this lab is to design and simulate a digital traffic light controller using a **finite state machine (FSM)**. Instead of depending on vehicle sensors, this controller follows a timed sequence that cycles the traffic lights for both streets. The goal is to show how digital logic, flip-flops, and timing signals work together to create predictable and safe light patterns.

In this lab, we examine different ways to design the controller, compare their complexity, and eventually build a version that will be implemented using real hardware in the next lab.

Part 1: Traffic Light Specifications and Design Choices

1. Sketch the intersection with the six traffic lamps. Label the street names and the R, G, and Y lamps for each direction.



2. Using the description above, create a table that shows the state of the six lamps at each of the 10 clock cycles. Since you do not know the states yet, just label each row 1–10.

Answer:

Cycle #	RNS	YNS	GNS	REW	YEW	GEW
1	0	0	1	1	0	0
2	0	0	1	1	0	0
3	0	0	1	1	0	0
4	0	0	1	1	0	0
5	0	1	0	1	0	0

6	1	0	0	0	0	1
7	1	0	0	0	0	1
8	1	0	0	0	0	1
9	1	0	0	0	0	1
10	1	0	0	0	1	0

Part 2: The 8-bit Approach

3. How many flip-flops (of any variety) are required to store the 8-bit state?

Answer:

We need a total of eight flip flops, each flip flop can store one bit and this design uses eight bits in total.

4. How many rows would your full transition table have?

Answer:

If we use eight flip flops, $2^8 = 256$, so we would need at least 256 rows for our transition table.

5. How many unused states are there?

Answer:

The traffic light only needs the ten states, but the system has 256 possible states, so 246 are left unused.

6. Imagine that when you power up your circuit, the starting state is random. What's the percentage chance that your controller misses your 10 used states? What dangers does this pose, and how would you try to fix it?

Answer:

There are 10 proper states out of 256 total states, so the chances of starting correctly is 3.9 %. We can fix this by adding a reset circuit that forces the system to start from a known state when powered on.

7. Considering the number of variables/combinations, do you think that designing an optimized circuit is practical?

Answer:

No, it's not practical. Using 8 bits creates way too many states and uses too many gates, and would use too many wiring. The circuit would be too large and it would be better to redesign it and optimize it.

Part 3: Minimal State Bits Approach, Version I

8. What's the minimum number of state bits required to support the 10 traffic controller states?

Answer:

The minimum number of state bits required would be 4 bits, because 2^4 is 16 and we only need 10 states which is more than enough.

9. How many rows will the transition table contain?

Answer:

We would need at least 16 rows for our transition table.

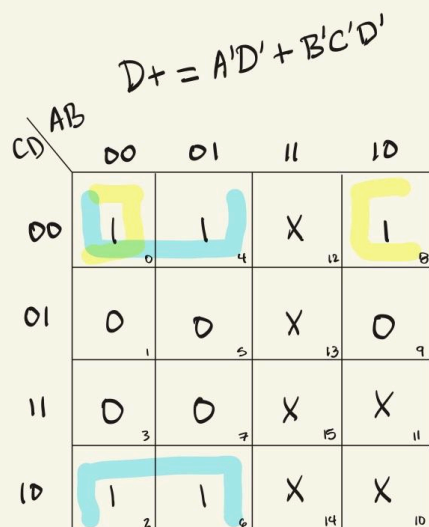
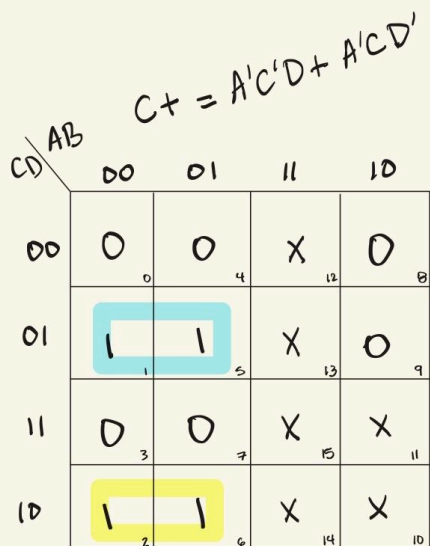
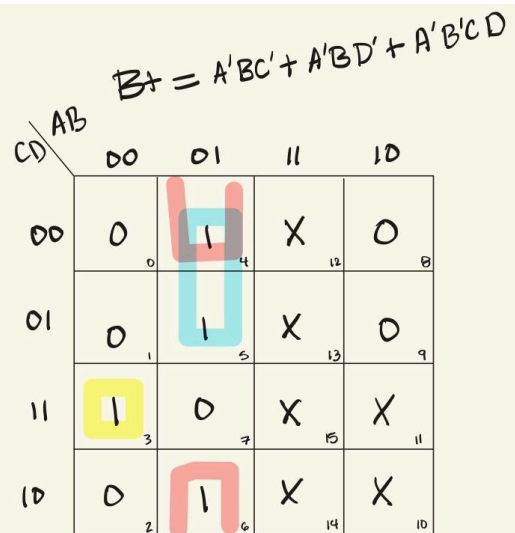
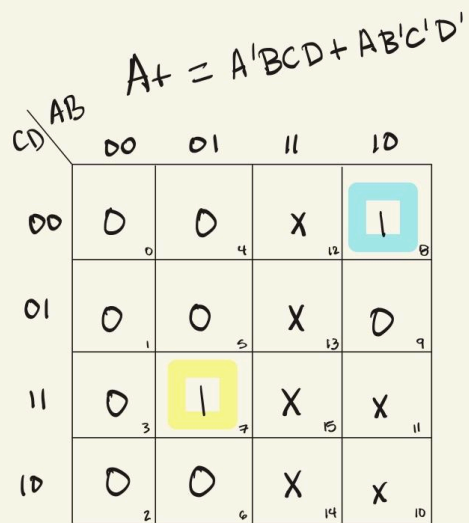
10. Create a 16-row transition table. Add columns for each of the six traffic light lamps as well. Leave unused states as don't-cares.

Answer:

Dec	A B C D	A ⁺ B ⁺ C ⁺ D ⁺	RNS	YNS	GNS	REW	YEW	GEW
0	0000	0001	0	0	1	1	0	0
1	0001	0010	0	0	1	1	0	0
2	0010	0011	0	0	1	1	0	0
3	0011	0100	0	0	1	1	0	0
4	0100	0101	0	1	0	1	0	0
5	0101	0110	1	0	0	0	0	1
6	0110	0111	1	0	0	0	0	1
7	0111	1000	1	0	0	0	0	1
8	1000	1001	1	0	0	0	0	1
9	1001	0	1	0	0	0	1	0
10	1010	XXXX	X	X	X	X	X	X
11	1011	XXXX	X	X	X	X	X	X
12	1100	XXXX	X	X	X	X	X	X
13	1101	XXXX	X	X	X	X	X	X
14	1110	XXXX	X	X	X	X	X	X
15	1111	XXXX	X	X	X	X	X	x

11. Using K-maps, solve for A⁺, B⁺, C⁺, D⁺, RNS, GNS, YNS, REW, GEW, and YEW as functions of A B C D given that YNS = 1 when A B C D = 0100. Show your work and place your formulas in a table.

Answer:



$RNS = A'BD + A'BC + AB'C'$

CD \ AB	00	01	11	10
00	0	0	X	1
01	0	1	X	1
11	0	1	X	X
10	0	1	X	X

$REW = A'B' + A'C'D'$

CD \ AB	00	01	11	10
00	1	1	X	0
01	1	0	X	0
11	1	0	X	X
10	1	0	X	X

$YNS = A'BC'D'$

CD \ AB	00	01	11	10
00	0	1	X	0
01	0	0	X	0
11	0	0	X	X
10	0	0	X	X

$YEW = ABC'D$

CD \ AB	00	01	11	10
00	0	0	X	0
01	0	0	X	1
11	0	0	X	X
10	0	0	X	X

$GNS = A'B'$

CD \ AB	00	01	11	10
00	1	0	X	0
01	1	0	X	0
11	1	0	X	X
10	1	0	X	X

$GEW = A'BD + A'BC + AB'C'D'$

CD \ AB	00	01	11	10
00	0	0	X	1
01	0	1	X	0
11	0	1	X	X
10	0	1	X	X

- Example:

- $A^+ = A'BCD + AB'C'D'$
- $B^+ = A'BC' + A'BD' + A'B'CD$
- $C^+ = A'C'D + A'CD'$
- $D^+ = A'D' + B'C'D'$
- $RNS = A'BD + A'BC + AB'C'$
- $YNS = A'BC'D'$
-
- $GNS = A'B'$
- $REW = A'B' + A'C'D'$
- $YEW = AB'C'D$
- $GEW = A'BD + A'BC + AB'C'D'$

Part 4: Minimal State Bits Approach, Version II

12. Create a 16-row transition table for your homemade states. Add additional columns for each of the six traffic light lamps. Leave unused rows as don't-cares.

Answer:

	Present ABCD	Next ABCD	RNS	YNS	GNS	REW	YEW	GEW
0	0000	0010	0	0	1	1	0	0
2	0010	0100	0	0	1	1	0	0
4	0100	0110	0	0	1	1	0	0
6	0110	0001	0	0	1	1	0	0
1	0001	1000	0	1	0	1	0	0
8	1000	1010	1	0	0	0	0	1
10	1010	1100	1	0	0	0	0	1
12	1100	1110	1	0	0	0	0	1
14	1110	1001	1	0	0	0	0	1
9	1001	0000	1	0	0	0	1	0
3	0011	XXXX	X	X	X	X	X	X
5	0101	XXXX	X	X	X	X	X	X
7	0111	XXXX	X	X	X	X	X	X
11	1011	XXXX	X	X	X	X	X	X
13	1101	XXXX	X	X	X	X	X	X
15	1111	XXXX	X	X	X	X	X	X

13. Explain (with words) how to find the state of GEW given any state in your sequence.

Answer: GEW is ON exactly when the sequence is on the "EW green" portion of the cycle.

GEW turns on when the East-West direction should have a green light.

In our design:

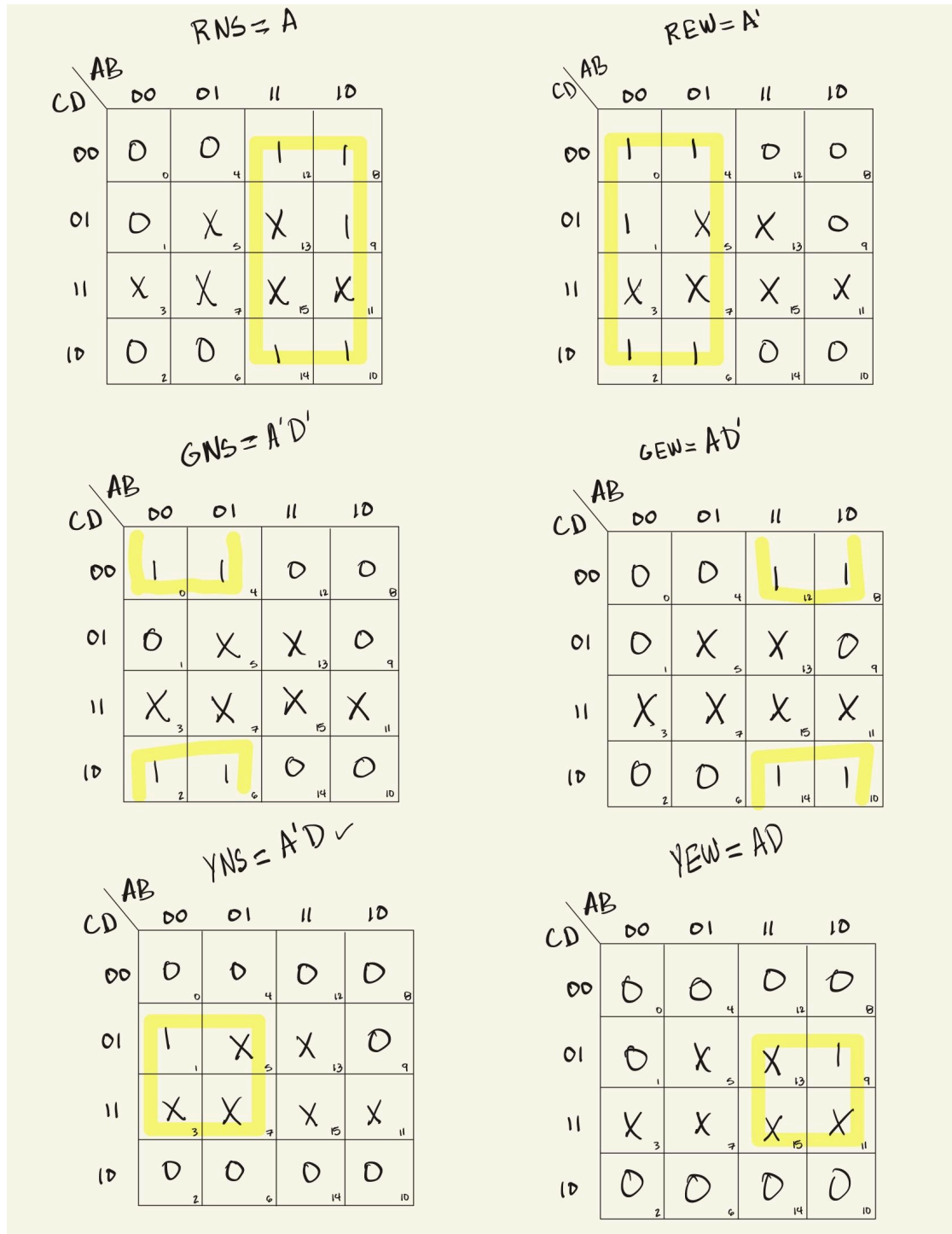
- A = 1 means it is EW's turn.
- D = 0 means the light is still green (not yellow).

During all EW green states in the table, A = 1 and D = 0.

During EW yellow, D becomes 1 so GEW must turn off.

$GEW = AD'$

14. Solve for A^+ , B^+ , C^+ , D^+ , RNS, GNS, YNS, REW, GEW, and YEW using K-maps. Show your work. Answer:



$$A+ = AD' + A'D$$

CD \ AB	00	01	11	10
00	0	0	1	1
01	1	X	X	0
11	X	X	X	X
10	0	0	1	1

$$B+ = B'C + BC'$$

CD \ AB	00	01	11	10
00	0	1	1	0
01	0	X	X	0
11	X	X	X	X
10	1	0	0	1

$$C+ = C'D'$$

CD \ AB	00	01	11	10
00	1	1	1	1
01	0	X	X	0
11	X	X	X	X
10	0	0	0	0

$$D+ = BC$$

CD \ AB	00	01	11	10
00	0	0	0	0
01	0	X	X	0
11	X	X	X	X
10	0	1	1	0

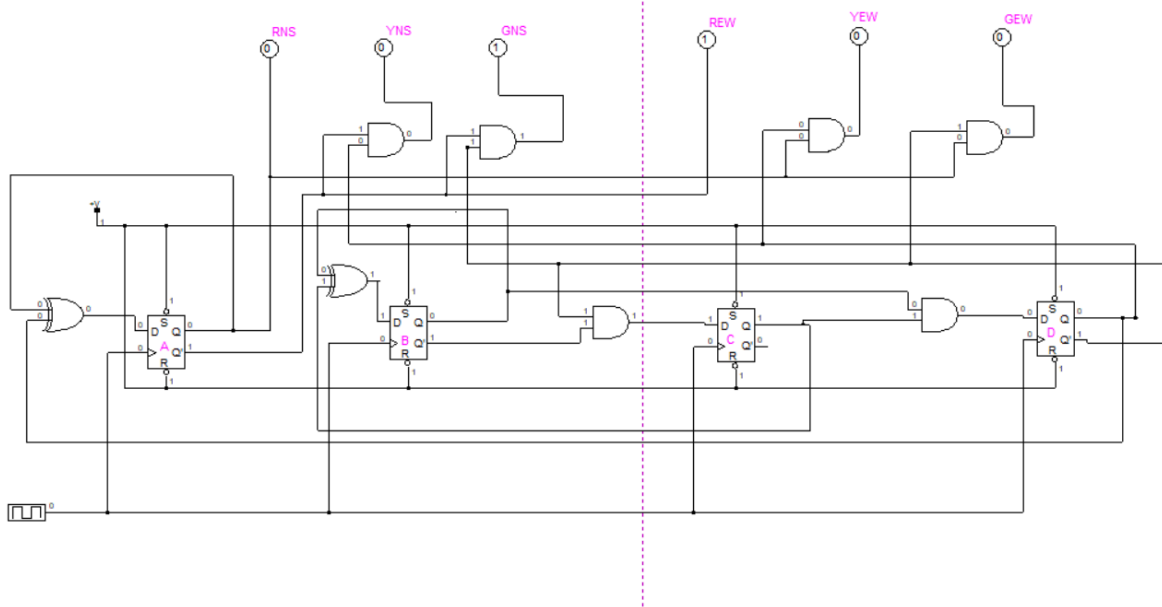
15. Does your Boolean algebra solution for GEW match your verbal explanation above?

Answer:

Yes, GEW is ON exactly during the portion of the traffic light cycle when EW direction has a green light

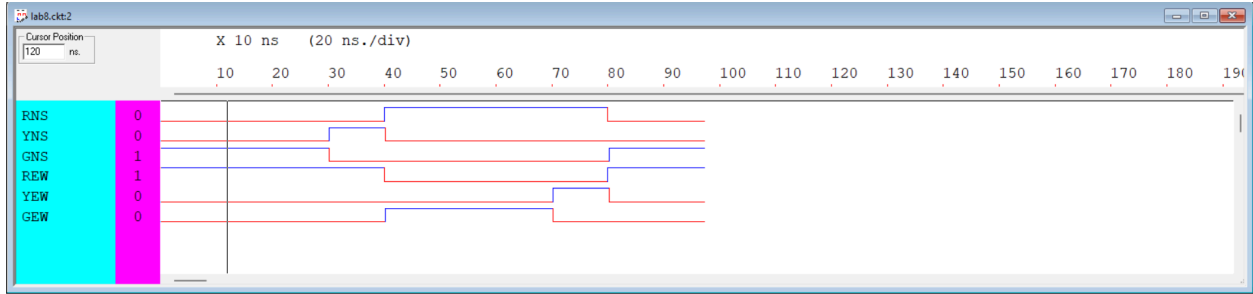
16. Provide a circuit schematic for your solution.

Answer:



17. Provide a timing diagram for ten clock cycles.

Answer:



18. Was your solution better than the one in Part 3? If not, try again! It will be worth it.

Answer:

My new solution used less gates and was more simple to make.

Conclusion

In this lab, we learned how to design a digital traffic light controller using a finite state machine. By comparing different design approaches, we saw how the choice of state sequence affects the complexity of the logic circuit. Using K-maps helped us simplify the next-state equations and the light-output equations. This project showed how digital systems can be built step by step, from understanding the problem, to designing the states, to creating real hardware using flip-flops and logic gates.